Deep Learning In Audio Ex2

Gal Barak (211699707), Ilia Benkovitch (316857820), Adam Fleisher (211603469)

December 2025

Question 2 Section d:

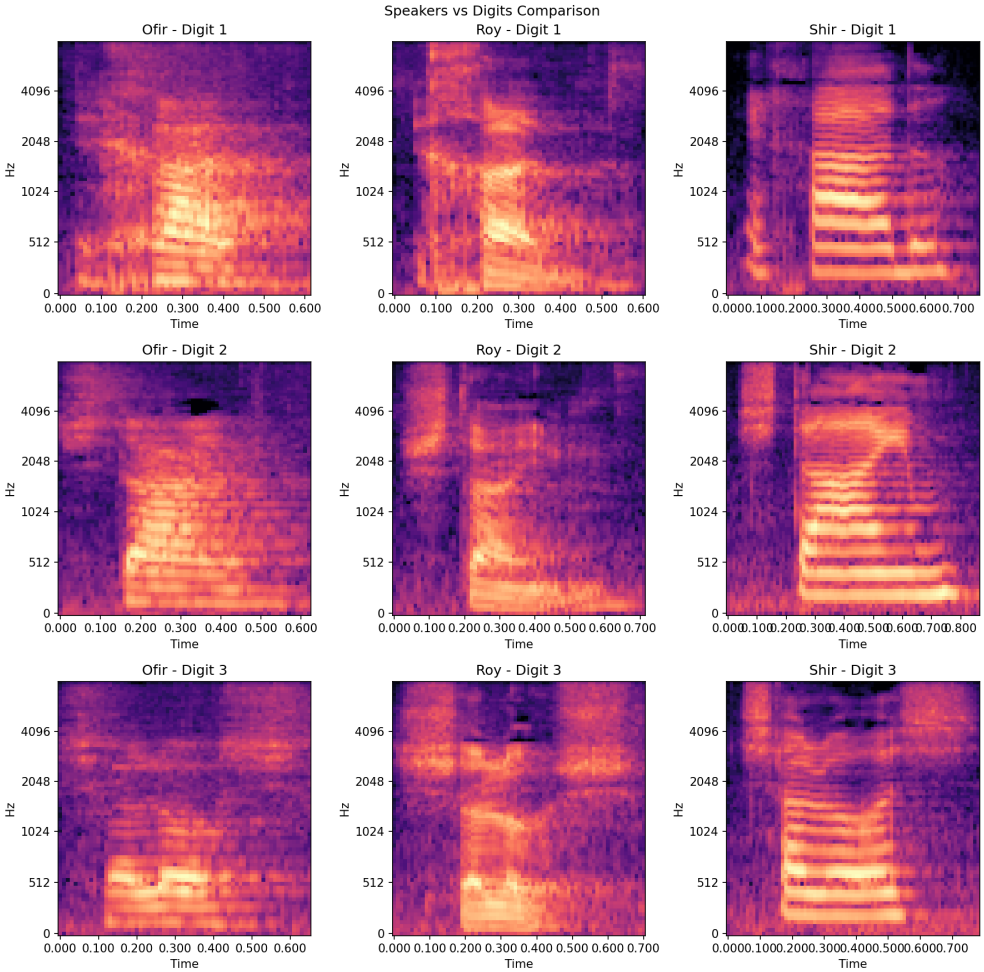

Figure 1: Ofir Vs Roy Vs Shir Mel Spectograms

Within the same speaker (Shir), the Mel spectrograms of different digits still look overall "Shir-like" — the general shape and energy pattern is similar. But you can clearly see differences between the words themselves. For example, שתיים and שלוש look more similar to each other than אחד, which makes sense because אחד doesn't start with ש.

You can also see the ש sound really clearly: in שלוש, at the beginning and the end there's mostly high-frequency energy, because ש is a "hissing/whistling" sound without vocal cords. In שתיים, you mainly see that high-frequency-only part only at the start (on the ש), not as much at the end.

When comparing speakers, you can notice that Shir's spectrograms generally have more energy in the higher frequencies, which often matches the fact women usually have higher pitch. Roy and Ofir look more similar to each other: their energy is spread in a closer way across bands, and the overall patterns across the same digits look more alike compared to Shir.

Question 3

a

We have chosen Gal's recordings as the reference database. Those can be found in the folder named "Samples/Segmented/Gal".

b

We have implented the algorithm and it can found in the file DTW under the function name dtw.

c+d

We have implemented this using the function "calc distance matrix" and choosing to not set a thresh (set thresh to None in the function plot distance matrices). We got the following results, where the red x marks the most fitting match for its respective column.

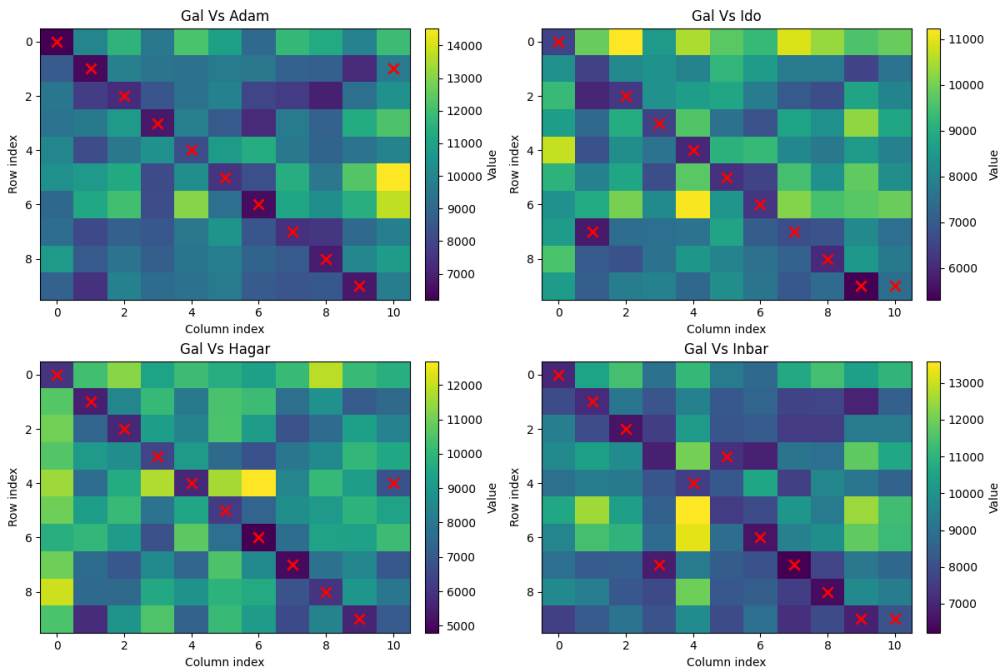

Figure 2: Distance matrix of the train set from Gal

e

We have chosen 6776 as our threshold since it is the highest possible threshold that doesn't yields incorrect results over the random word. We have gotten the following results:

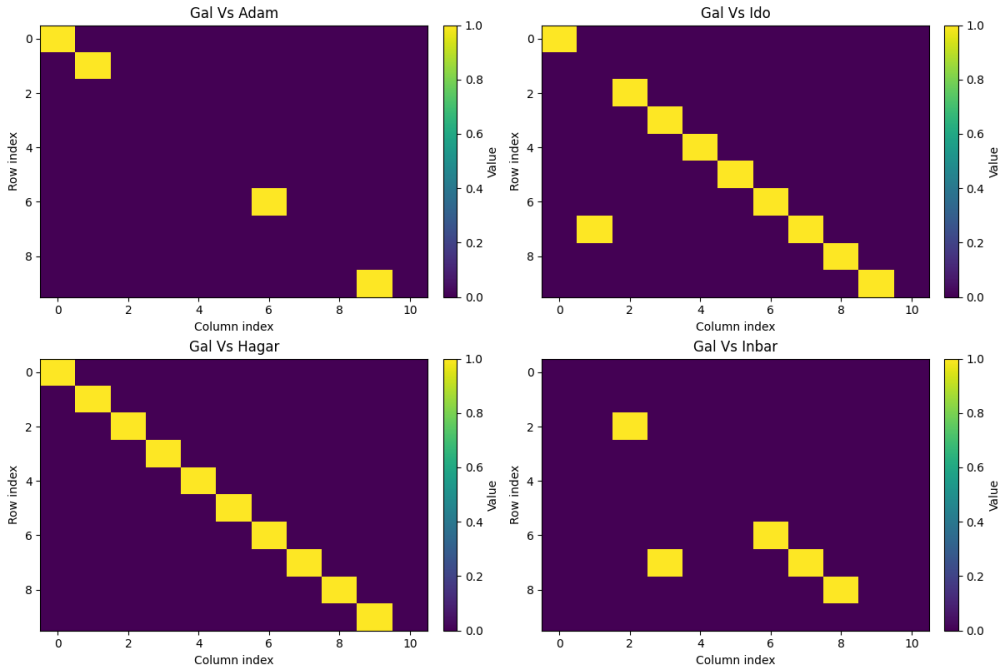

Figure 3: Classification Results After Threshold

f

We have gotten the following confusion matrix with an accuracy of 67.5%:

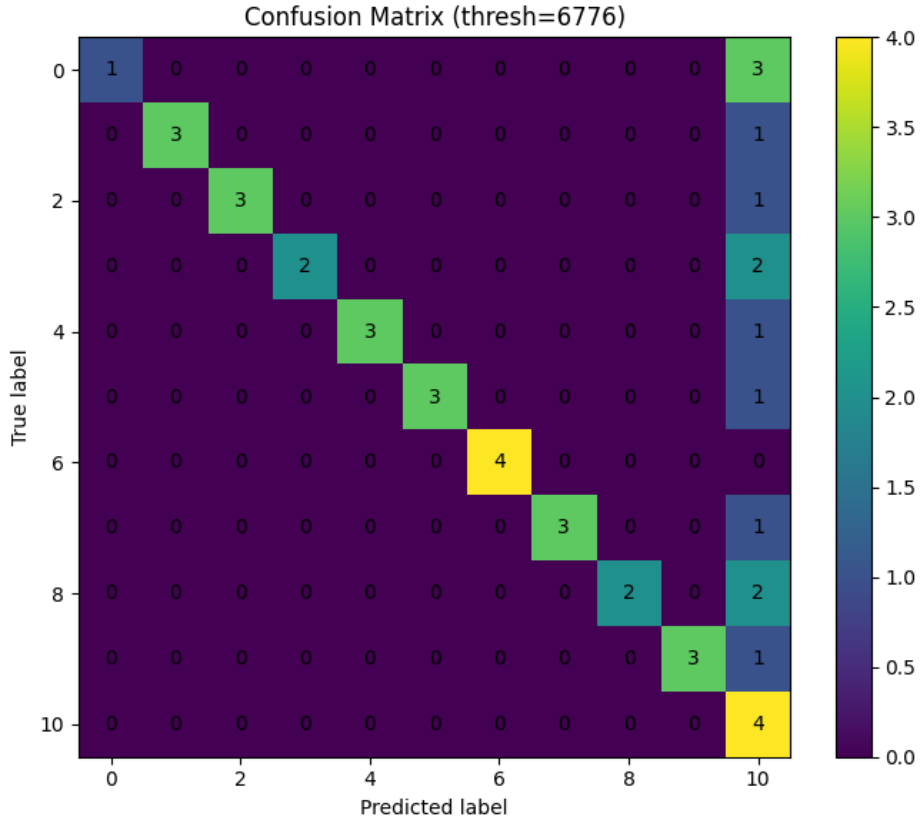

Figure 4: Confusion Matrix Over Validation

g

We have implemented both AGC normalization and length normalization and we tried finding the optimal solution over all 4 combinations:

1. No normalization
2. Only length normalization
3. AGC normalization
4. Both length and AGC normalization

We have found that we got the best results with length normalization (without AGC) and when setting the threshold to 59. We got an accuracy of 97.7% (Which even surprised us :)). Using both types of normalization we got almost the same results (95.5% accuracy). This is the confusion matrix we got using only the length normalization:

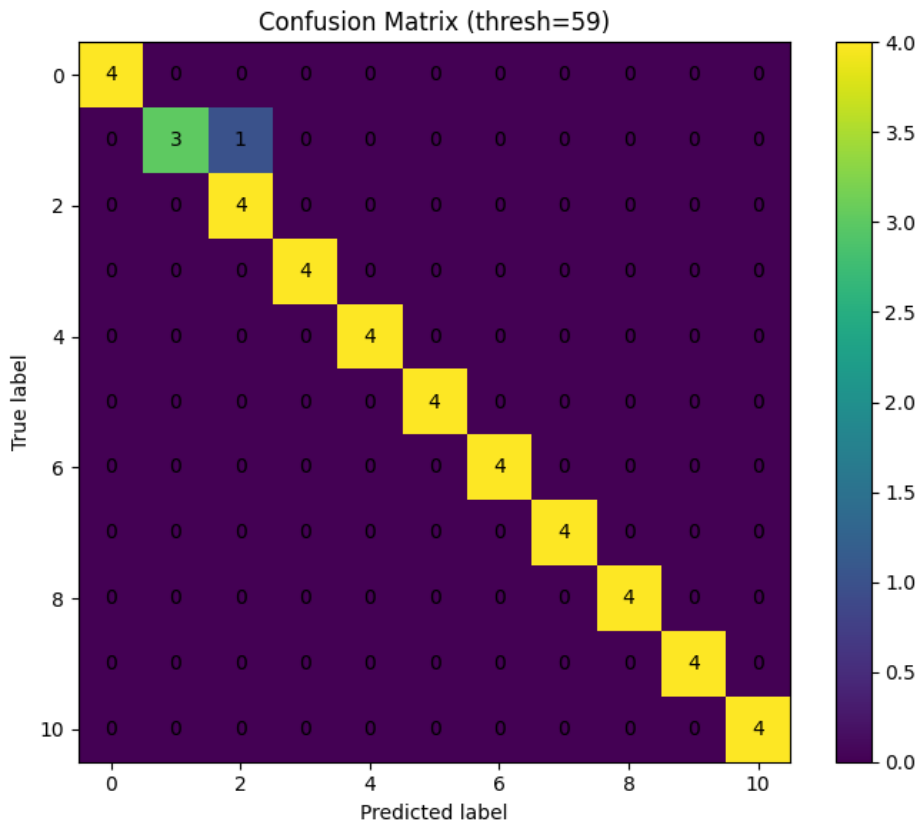

Figure 5: Confusion Matrix With Length Normalization

Question 5

c

Given the provided probability matrix, the probability to get the sequence 'aba' is: $Pr('aba') = 0.0888$

d

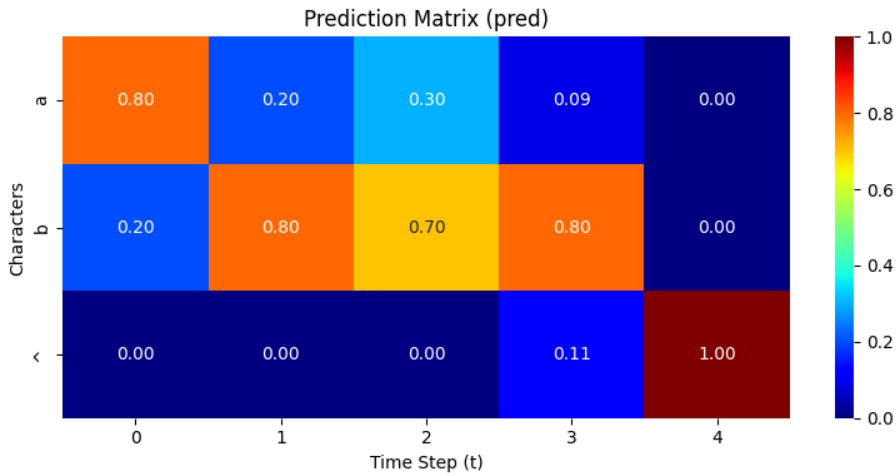

Figure 6: Pred matrix

Question 6

b + c

The most probable path for the sequence 'aba' is: `abba^`, with probability 0.0403 (using *max* instead of *sum*)

d

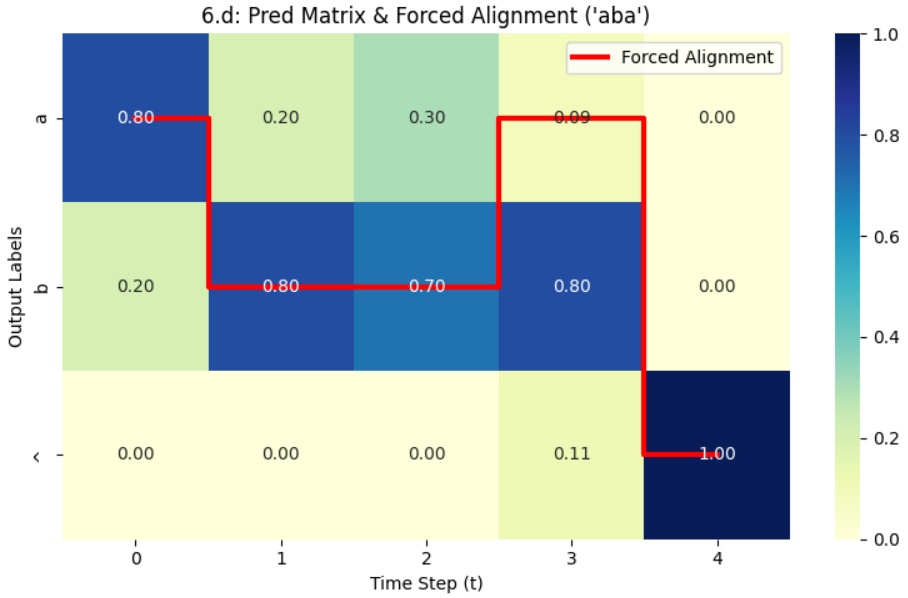

Figure 7: Aligned Sequence

e

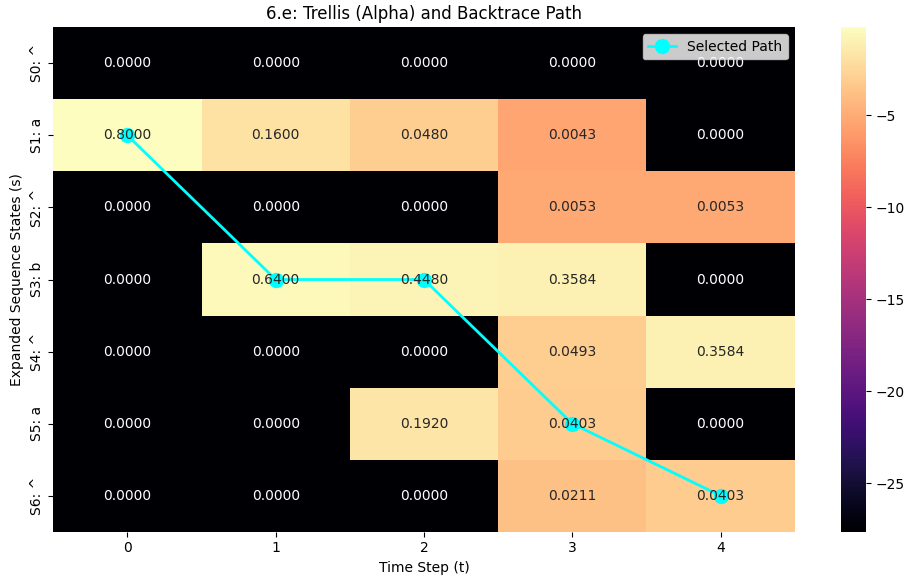

Figure 8: Backtrace Matrix

Question 7

b + c

Probability to get 'then goodbye said the rats and they went home' with forced alignment (max instead of sum) = 0.00001030

d

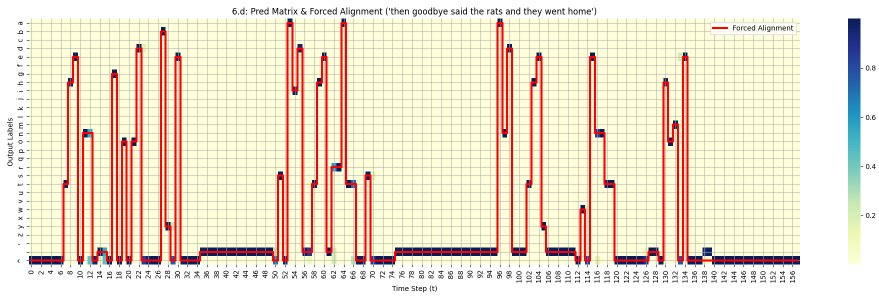

Figure 9: Aligned Sequence

e

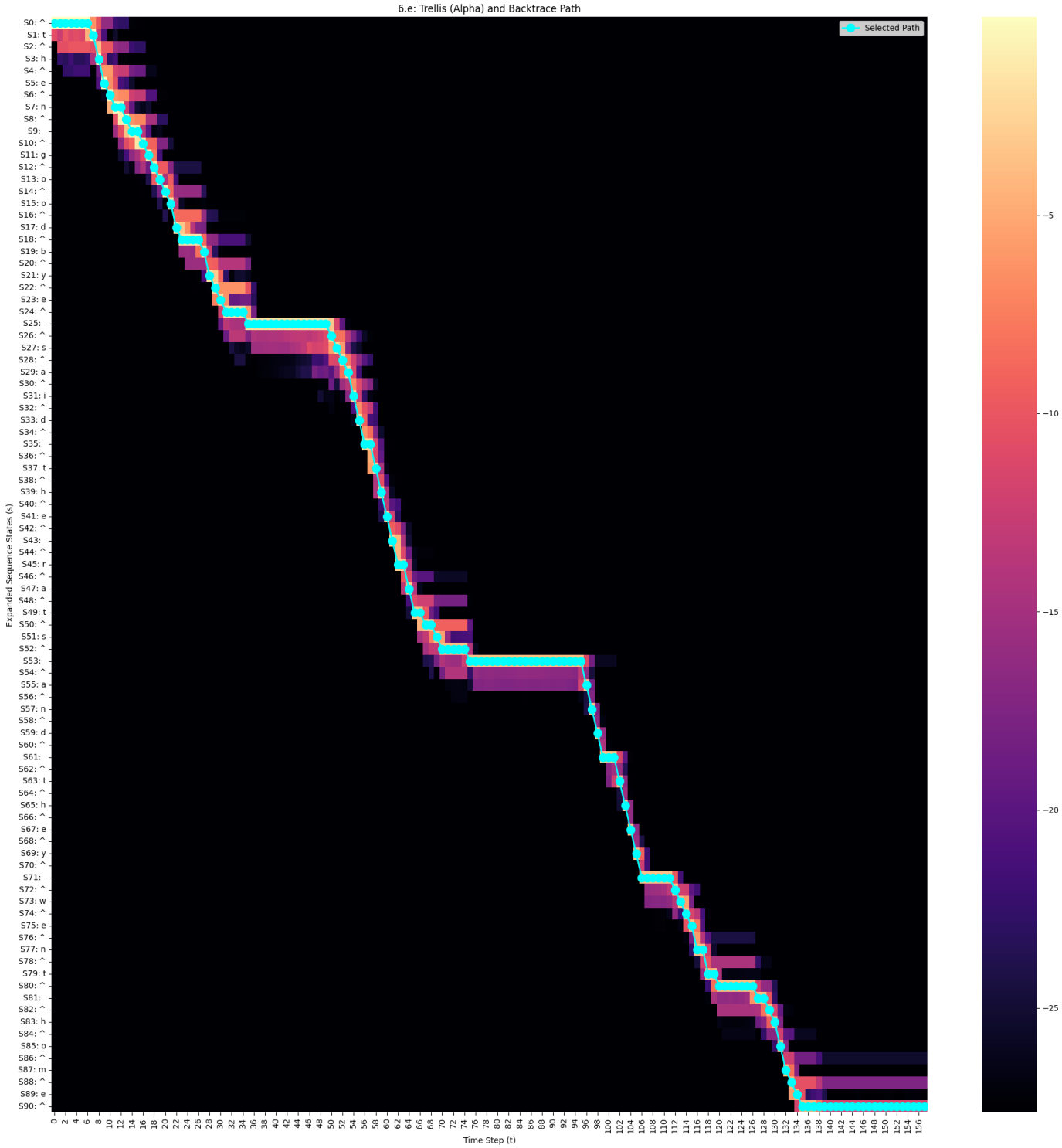

Figure 10: Backtrace Matrix